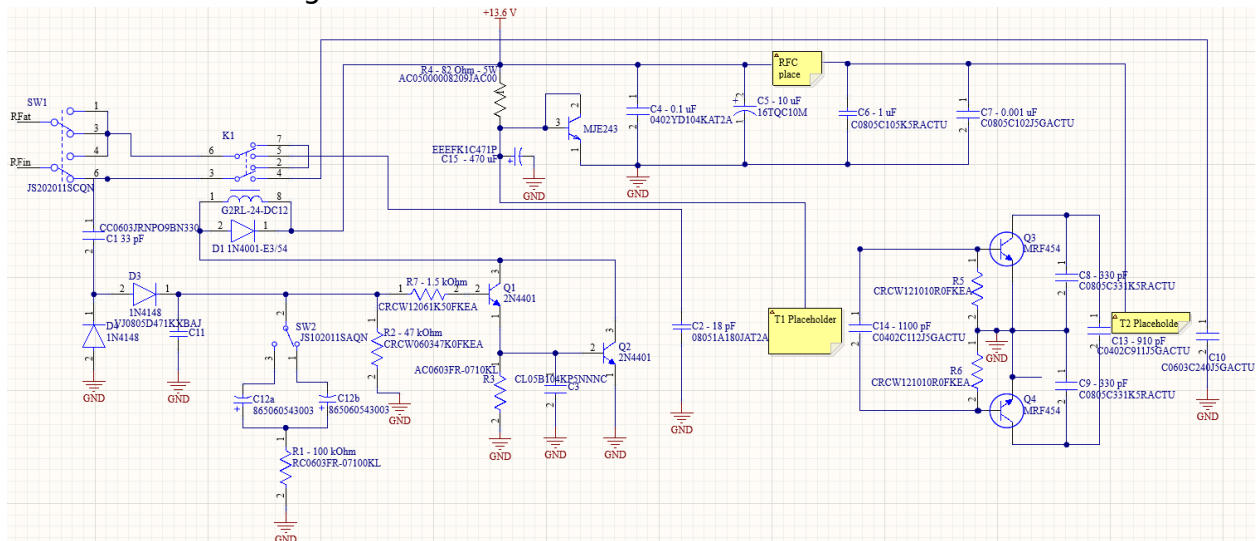


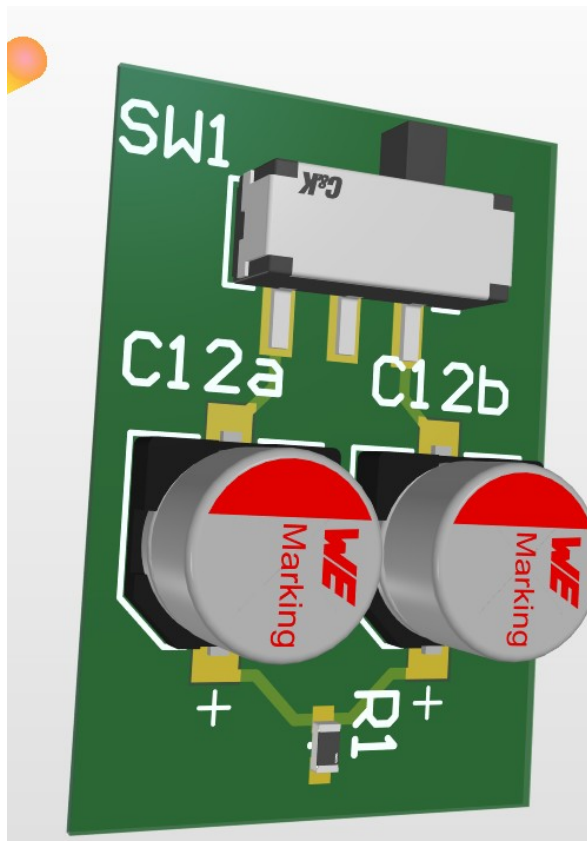
PCB Process

The first board: EB63A. However we decided to separate the EB63A board with the Switch board starting at SW2 with C12a and C12b.



Switch board:

First revision. Decided going with all of the small part common for PCB.



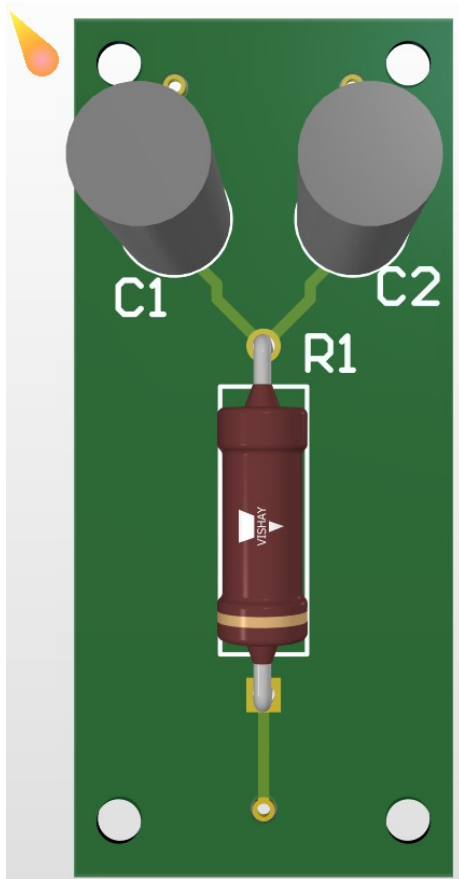
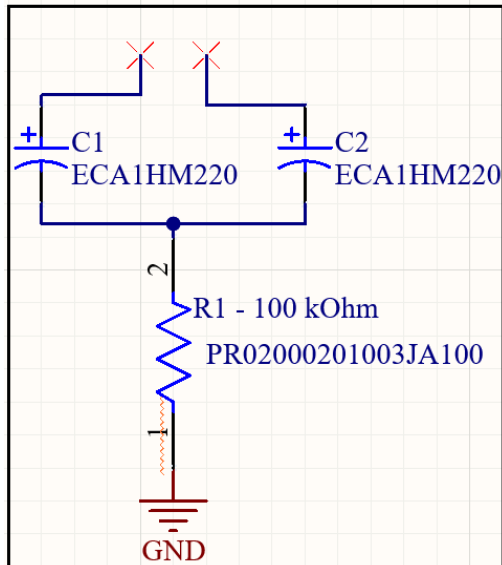
However we later decided to go with regular size switches and resistor for:
Easier and cheaper to buy/Install

PCB Process

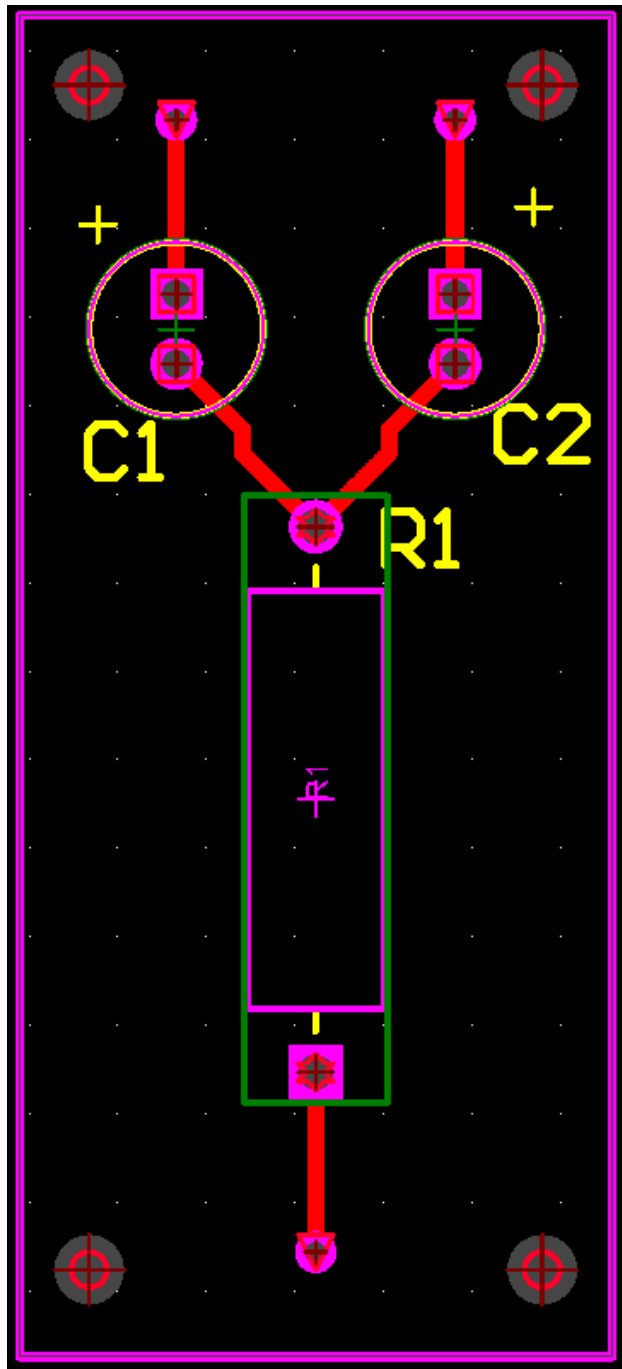
Switches can actually be shown on the enclosure for demonstration purposes

Final version: The layout of the PCB and the circuit

C12a/C12b Switch Connector



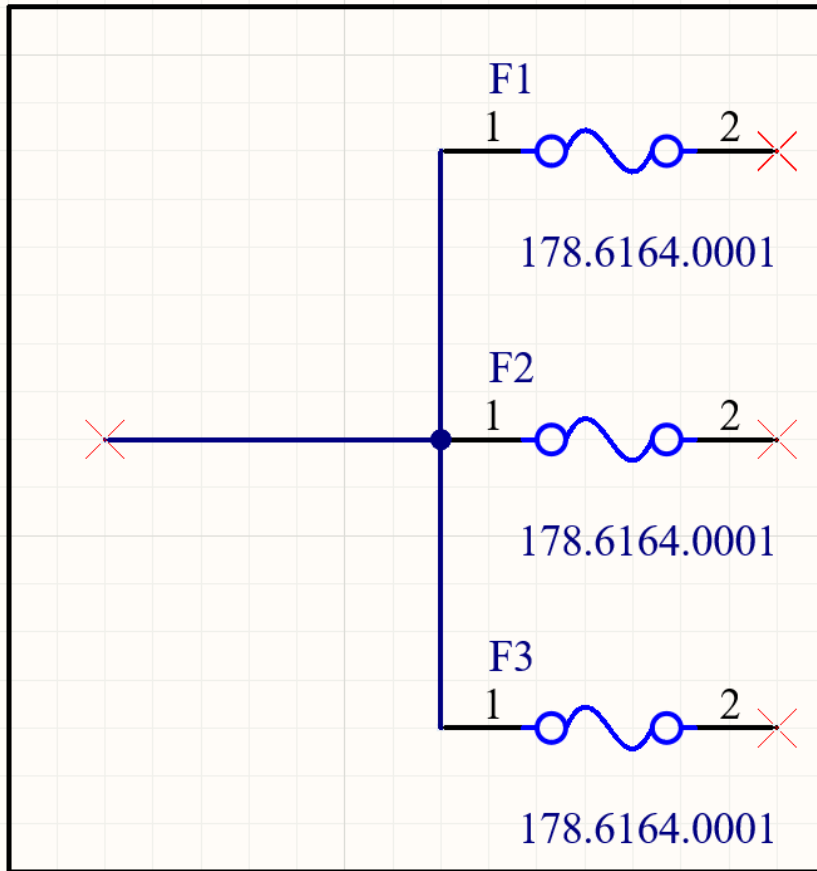
Below is the printed circuit board for laser cutting



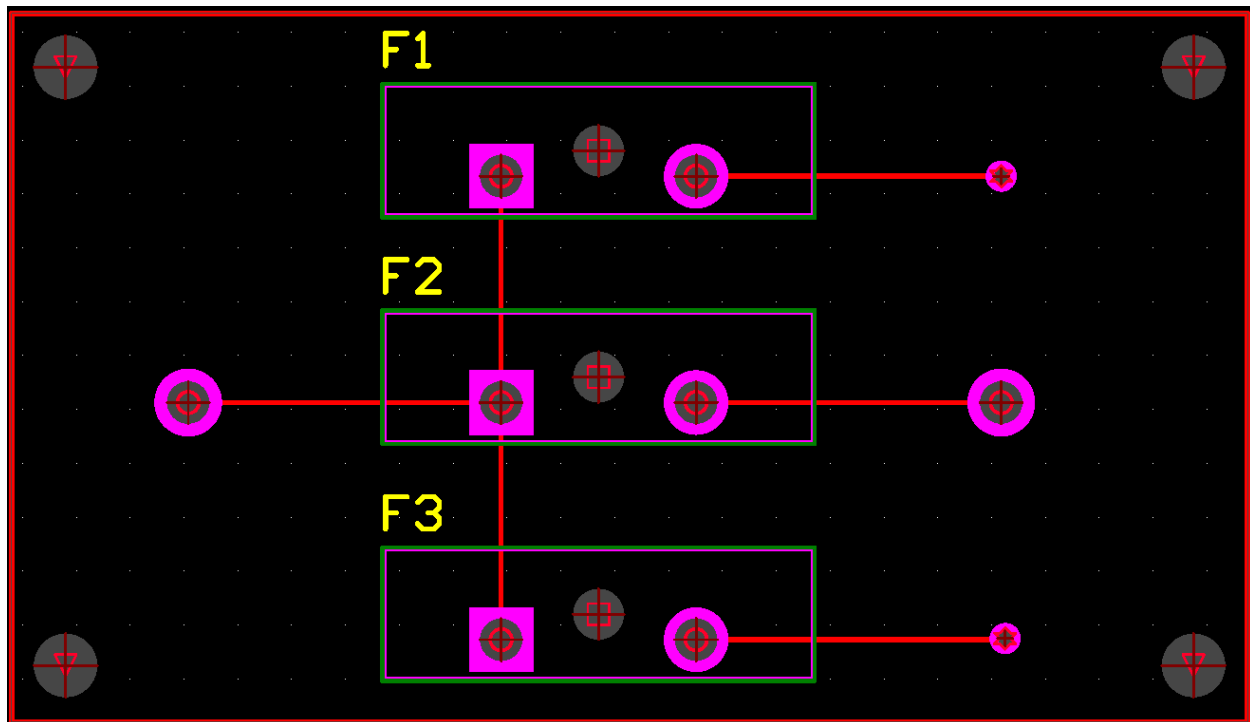
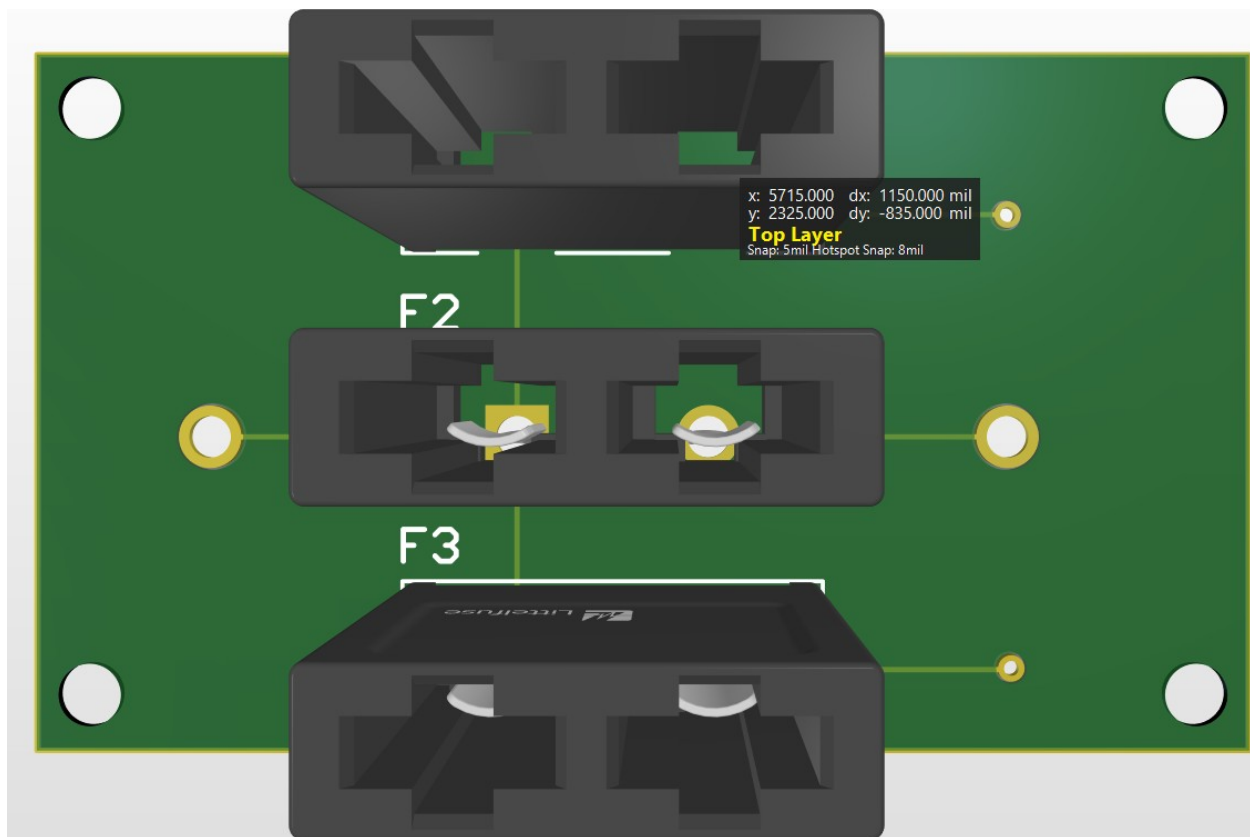
The fuse board:

Final version with the Circuit/PCB implementation and Layout for laser cutting.

Fuse Board



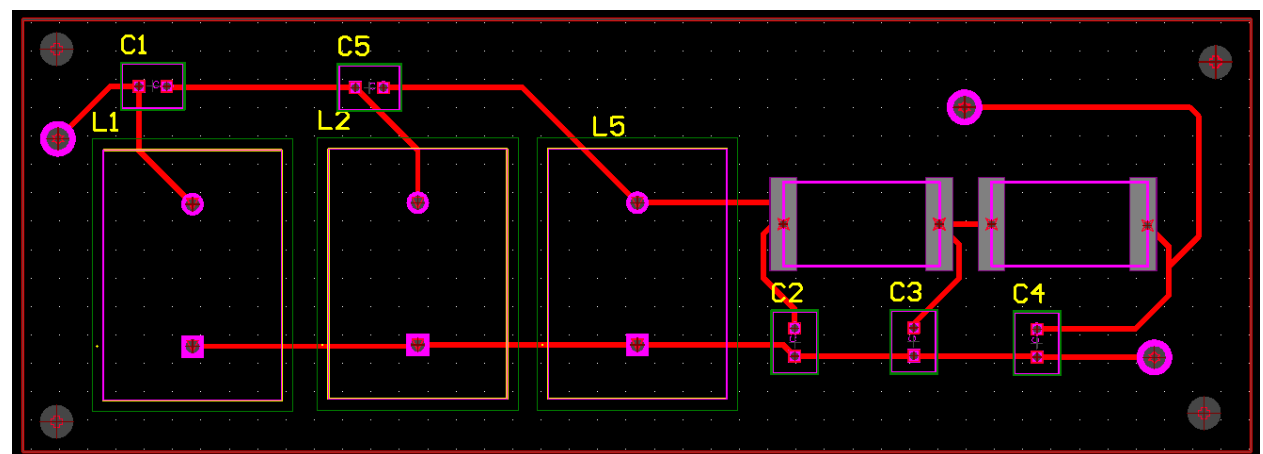
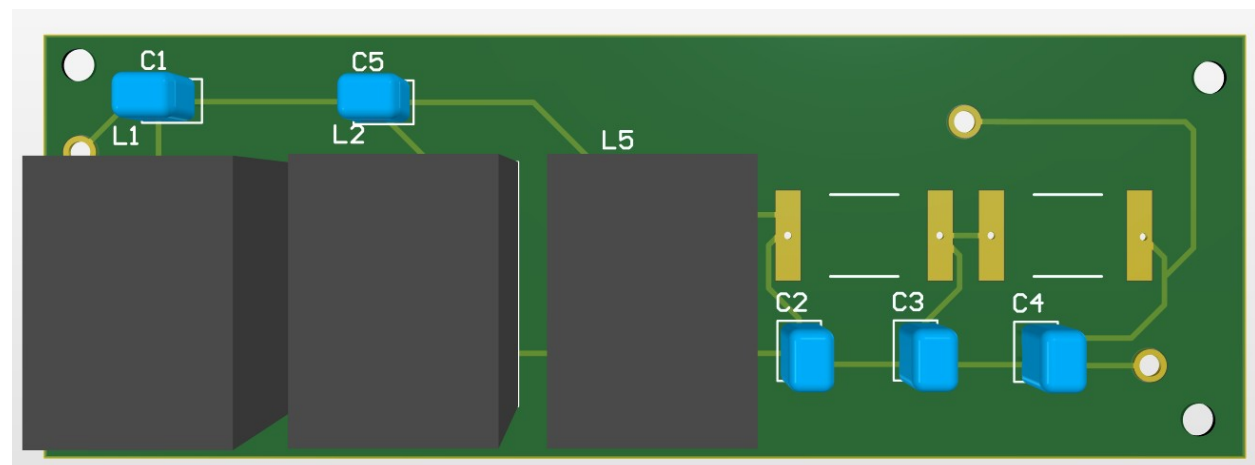
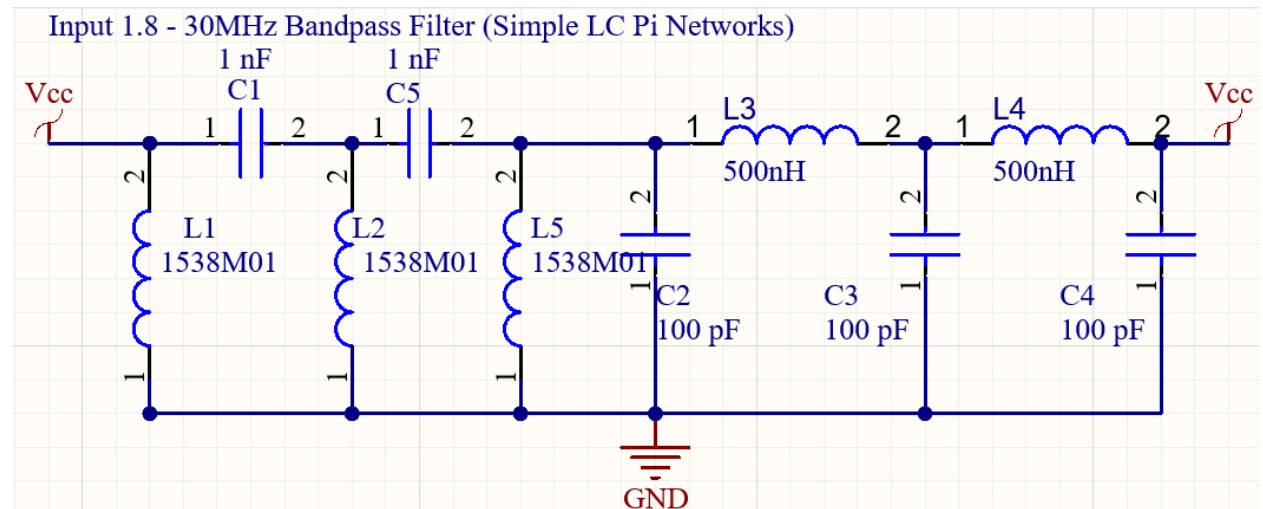
PCB Process



Bandpass Filter PCB:

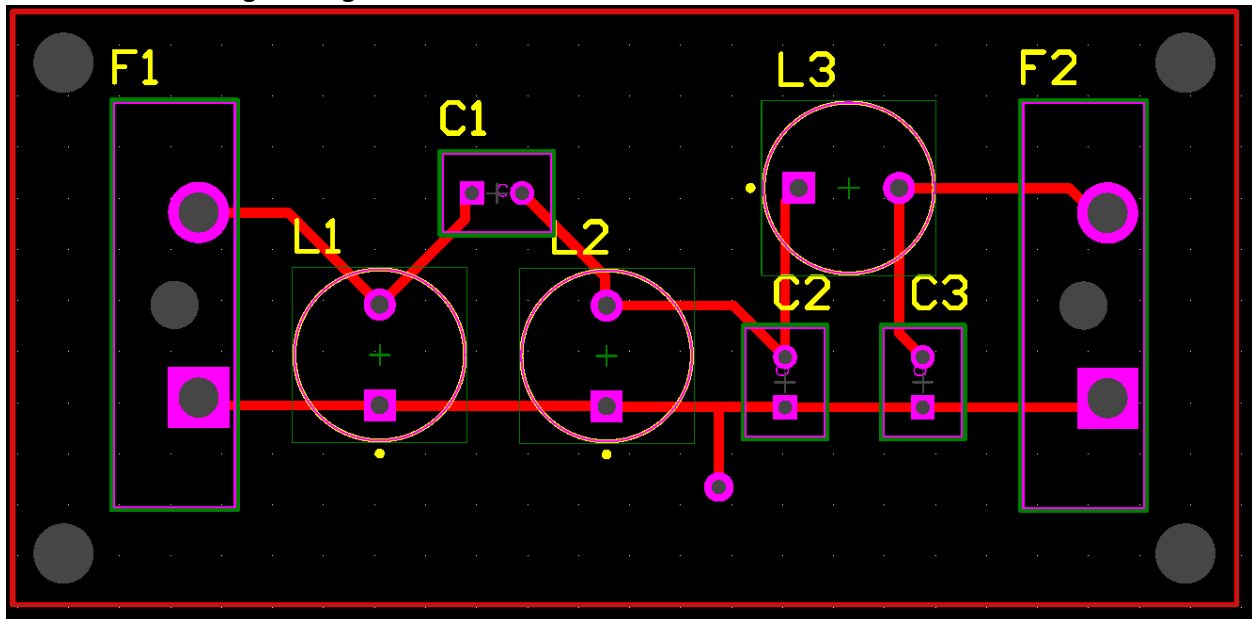
The final circuit and PCB layout

PCB Process

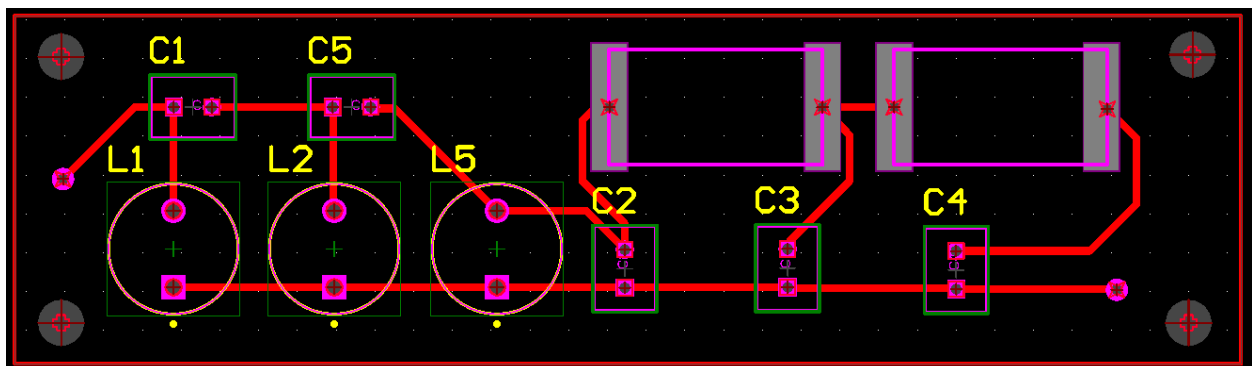


PCB Process

Version 1: Wrong configurations

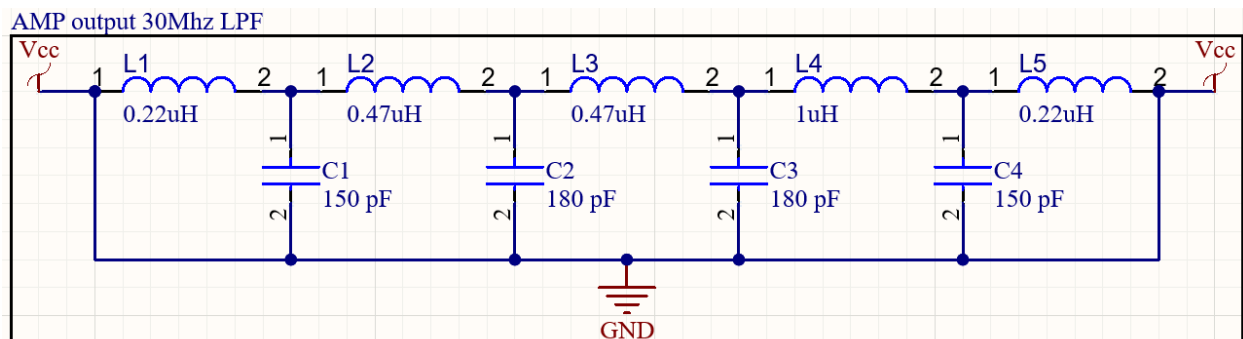


Version 2: Using the wrong capacitors and not having an output.

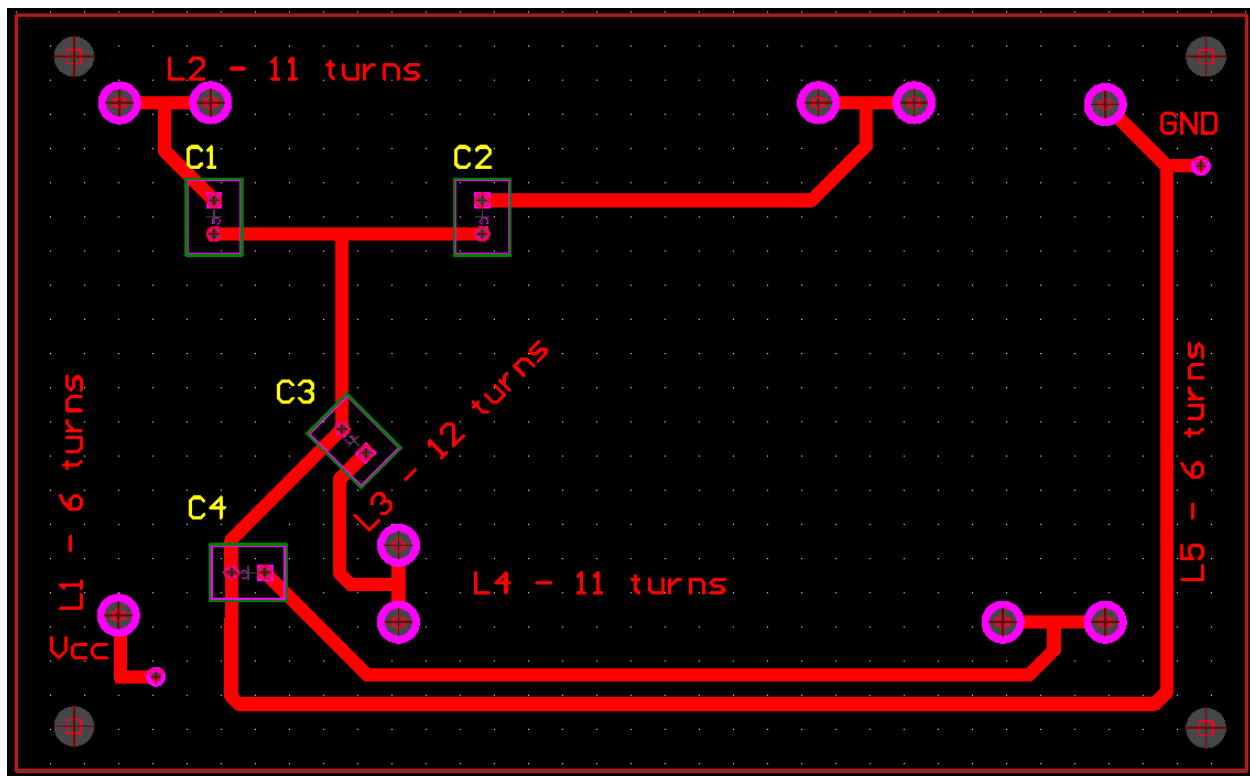
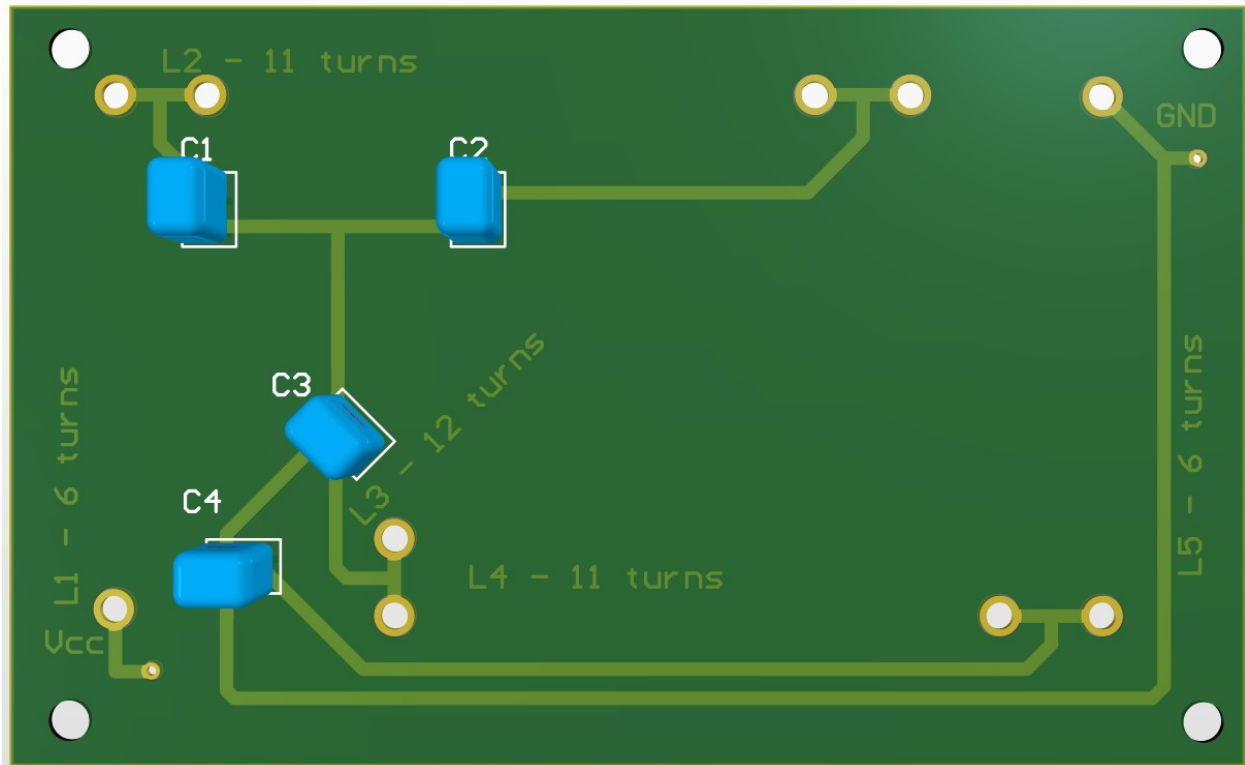


AMP Output Board:

Final version:

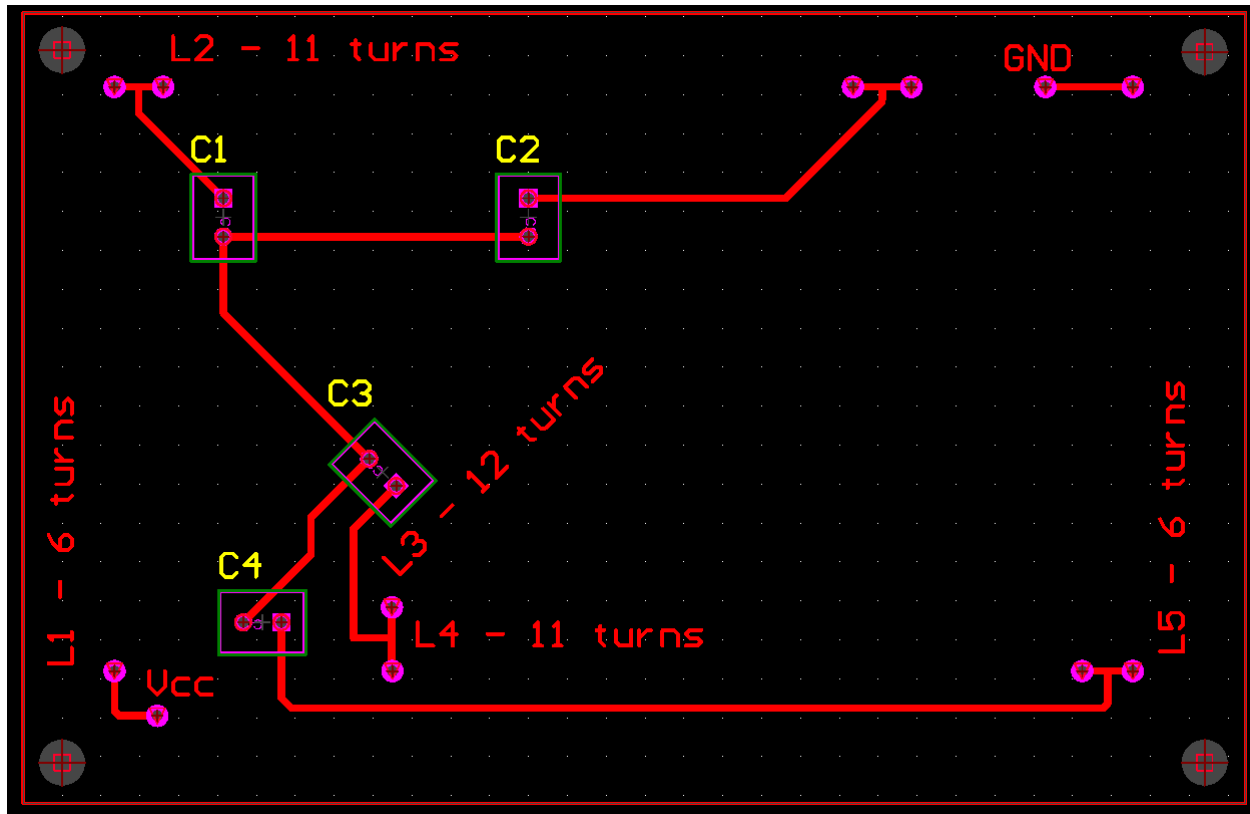


PCB Process



Version 1: As this board will generate a lot of current, we decided to increase the traces from 0.5 mm to 0.8~1 mm.

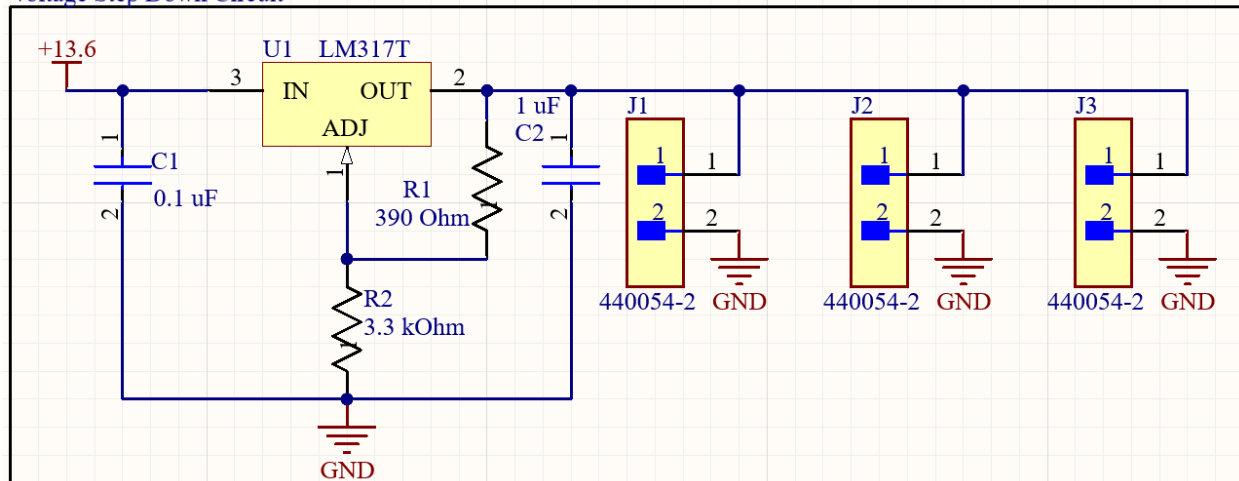
PCB Process



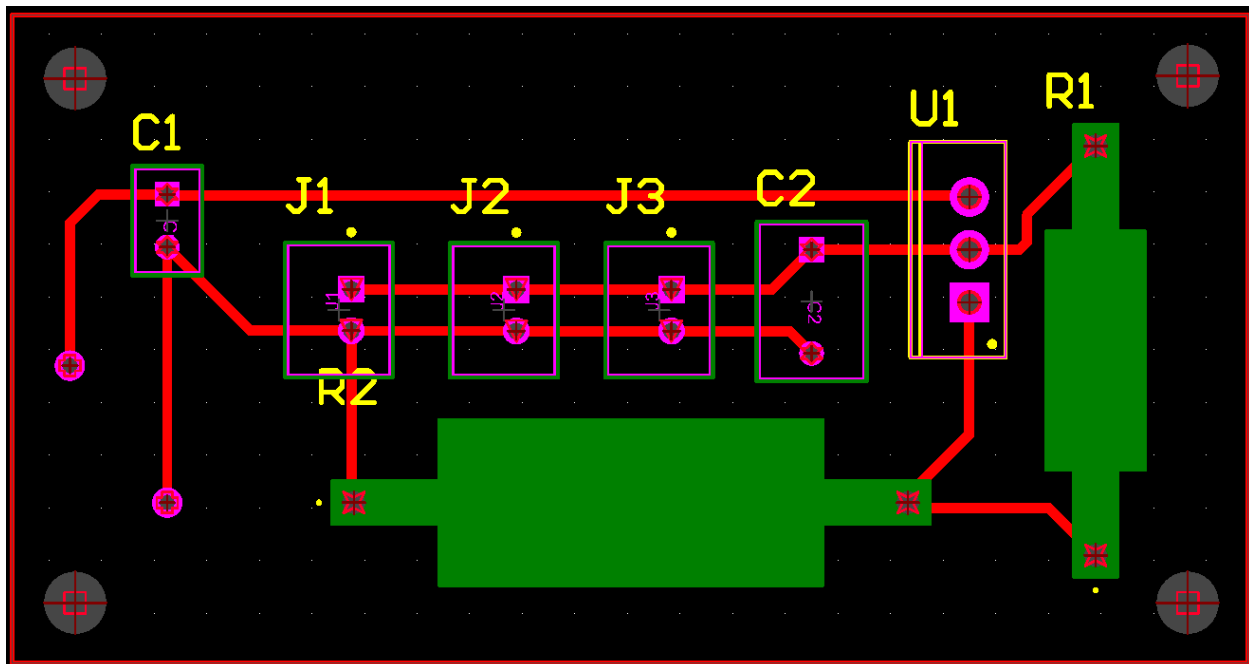
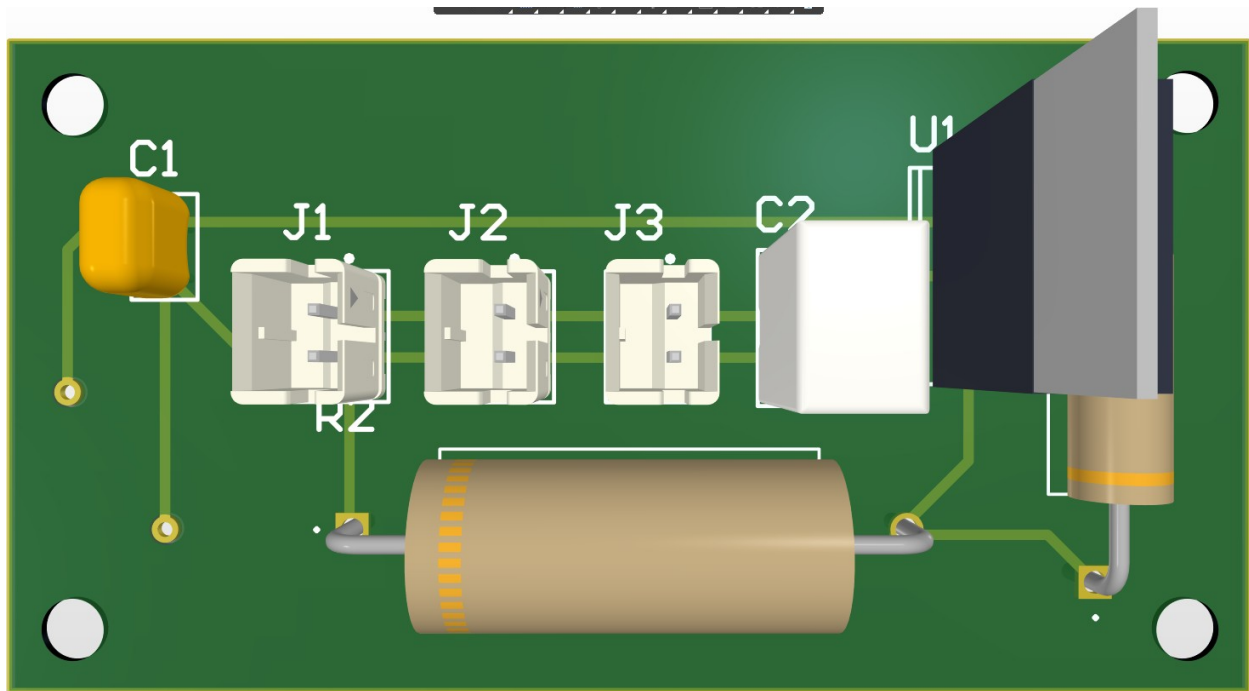
Voltage Stepdown circuit PCB:

Final Version:

Voltage Step Down Circuit



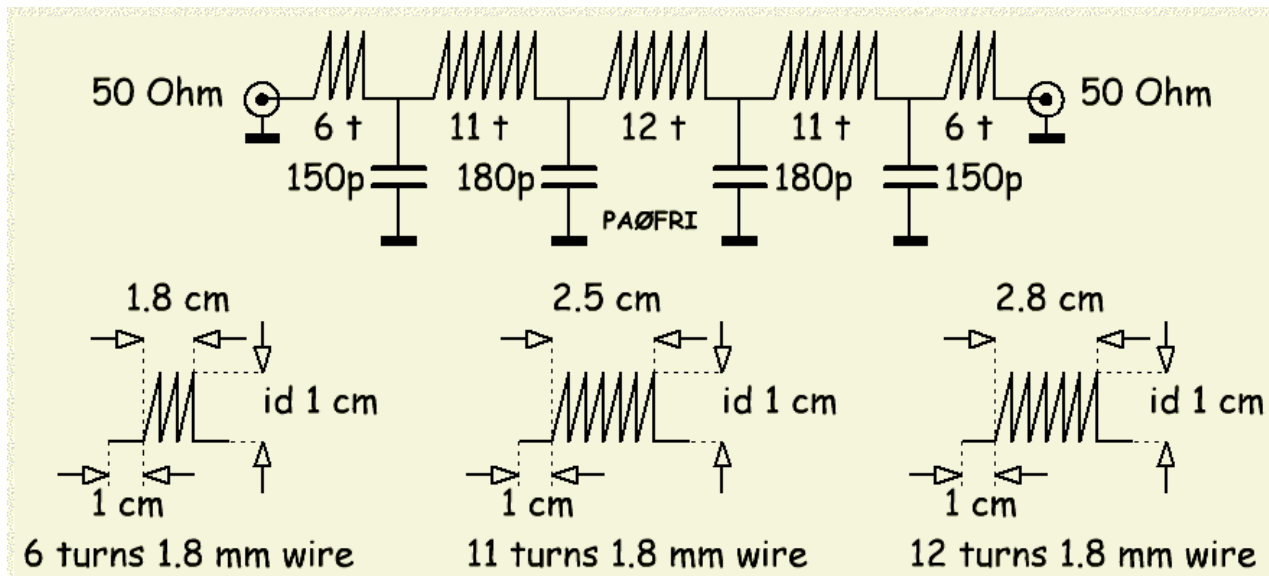
PCB Process



Note:

All of the circuit was hand drawn at first and later implemented into Altium Designer. Most of the selected components are according to what will be used. However, there are place holders (such as transformer coils) that will be custom made by us.

PCB Process



There are minor changes during the PCB processes such as: Changing I/O holes size according to the wire gauge we got,...But these are minor details not worth mentioning.

Difficulty during the PCB design process:

Getting used to Altium Designer was difficult at first since we only accustomed to KiCAD. However after I pass the initial learning curve, Altium Designer is way more powerful and more customizable.

Not all of the required components are available on PCB. So there are some parts that we have to implement/imported either by Mouser's 3D models or custom made.